

MME8215: Operations Research - Seminar

II. Linear Programming cont.

Propose a model for solving the following problems. When applicable solve the problem graphically and algorithmically (try to draw the feasible solution space and solve the problem with simplex and/or using your implementation or an appropriate API).

1. Production Mix Production Mix A company produces two kinds of products, $P1$ and $P2$. The selling price of $P1$ is 300€ The selling price of $P2$ is 250€

- $P1$ requires 2 operator man-hours and 1 hour of machine operating time.
- $P2$ requires 1 operator man-hours and 3 hours of machine time.
- Operator man-hours cannot be higher than 40 hours per day.
- Machine operating time cannot be higher than 45 hours per day.
- Marketing states that $P1$ units per day cannot be higher than 12.

How many units of $P1$ and $P2$ should the company produce every day to maximise income?

2. Ski Patrol Ski Patrol Ltd. manufactures two different models of snowboards branded Normal and Deluxe with a weekly production plan and inventory. They sell their material to local shops. The company has a profit of 300€ per Normal snowboard and 400€ per Deluxe snowboard. The manufacturing process consists of roughing and polishing.

The Normal snowboard model requires 20 minutes of roughing and $\frac{2}{3}$ minutes of polishing. The Deluxe snowboard model requires 40 minutes of roughing and 1 minute of polishing. The roughing equipment is available for 1000 minutes per week and the polishing equipment for 800 minutes per week. Ski Patrol has already signed a contract with a local shop called Downhill Racer to deliver 18 Normal snowboards per week.

Ski Patrol hires you to formulate a Linear Programming Problem to help them determine the best production mix to maximise profits.

3. The good carpenter “The Fog Ltd” is a modern furniture company that makes tables and chairs. The selling price of a table is 30€ and the selling price of chairs is 10€. The carpenter at the company works up to 40 hours and needs 6 hours to make a table and 3 hours to make a chair. Customer demand requires that he makes at least 3 times as many chairs as tables. Tables take up four times as much storage space as chairs and there is room for at most four tables each week.

What is the optimal weekly production plan that optimises revenues? Formulate this problem as a linear programming problem.

4. Blending Problem The Kahuna company manufactures sausages using three kinds of meat. The relevant information about the ingredients is provided in the table below:

Ingredient	Cost (€/kg)	Availability (kg)
Pork	4.32	30€
Wheat	2.46	20€
Starch	1.86	17€

The company makes two types of sausages:

- Economy ($\geq 40\%$ Pork)
- Premium ($\geq 60\%$ Pork)

One sausage is 50 grams (0.05 kg). According to government regulations, the most starch that can be used in sausages is 25%. The company has a contract with a butcher, and have already purchased 23 kg pork, that will go bad if it's not used. The company has a demand for 350 economy sausages and 500 premium sausages.

Write a linear program to figure out the most cost-effective recipe to manufacture our sausages.

5. Petroleum Blending Harry Stamper is an engineer working on a petroleum company. The company produces three grades of motor oil – *super*, *premium*, and *extra*. The three grades are made from the same 3 ingredients, named component 1, component 2 and component 3. The company wants to determine the optimal mix of the 3 components in each grade of motor oil that will maximize profit.

The maximum availability of each component and their cost per barrel are as follows: To ensure

Component	Max barrels available / day	Cost / barrel
1	4500	12€
2	2700	10€
3	3500	14€

the appropriate blend, each grade must have a minimum amount of component 1 plus a combination of other components as follows:

Grade	Component 1	Component 2	Component 3	Selling price/barrel
Super	$\geq 50\%$	$\leq 30\%$	-	23€
Premium	$\geq 40\%$	-	$\leq 25\%$	20€
Extra	$\geq 60\%$	$\geq 10\%$	-	18€

The company wants to make at least 3000 barrels of each blend. Formulate an LP to help Harry Stamper find the optimal blend that maximises profit.

Analysis questions (Do this part once you have reviewed LP analysis including sensitivity analysis questions)

- What is the total profit obtained by the company with the Extra grade with the solution provided by the solver?
- What is the maximum value that the profit per barrel of component 2, in the Extra grade, can take without changing the base of the optimal solution?

- If the cost of the barrel of component 3 is increased by 1€, will the optimal composition change? Motivate your response
- Do we use every barrel available of every component in the optimal solution? Or do we have any surplus availability for any component? Motivate your response
- The provider of Component 2 may have a problem with the delivery and the availability may be reduced to 400 barrels. Will this shortage affect the optimal composition? Motivate your response
- Grace wants to use the results to renegotiate the terms and conditions with the suppliers of the different components. Which component is a better candidate to increase the availability? How would you use the model to negotiate new terms with the provider, assuming you do not want any change in the base of the optimal solution? Motivate your response

6. All gold canyon A gold seeker needs to find at least 2 kg of bronze and 3 kg of copper to pay his debts. There are two rivers in which the miner can find these precious metals. Each day the miner spends in river *A*, he finds 0.3 kg of bronze and 0.3 kg of copper while each day he spends in river *B*, he finds 0.2 kg of bronze and 0.4 kg of copper.

Can you help this gold seeker to pay his debts as soon as possible?

7. Making weapons Din Djarin is a retired bounty hunter that now runs a small factory specialised in making classic weapons in a colony recently established on Mandalore. The factory manufactures two different types of weapons, swords and spears. Din uses special alloys, mixtures of metals and rare materials, which have made a name for the endurance of his weapons across the colony. The table below shows the weight in pounds of each material needed to make each type of weapon:

Metal	Spear	Sword
Beskar	1	0.5€
Tungsten	1	1€
Titanium	3	1€

The total amount of each metal available, also in pounds, is: Beskar - 125, Tungsten - 225, Titanium - 300. The price of a spear is 40 credits and the price of a sword is 5 credits. Din has already sold 100 swords, so he needs to manufacture at least that many.

Write a LP model to find the optimal number of spears and swords that maximise revenue.

8. Chappie Chappie Ltd. has 3 production lines (A, B, and C) to make 4 different types of metal sheets of width 0.4, 0.5, 0.6, and 0.7 mm. The company needs to meet the monthly demand for each kind of product (in tons) expressed in the next table:

Sheet 0.4 mm	Sheet 0.5 mm	Sheet 0.6 mm	Sheet 0.7 mm
500	1200	1500	300

The production lines do not have the same overall efficiency for all types of sheets. Furthermore, due to design constraints, not all the lines are capable of manufacturing all types of sheets. The table below indicates the manufacturing costs in € of the different products in the different lines: a dash “-” means that it is not possible to manufacture a product in a line:

Line	Sheet 0.4 mm	Sheet 0.5 mm	Sheet 0.6 mm	Sheet 0.7 mm
A	60	50	50	45
B	80	70	75	70
C	-	60	60	-

The different lines have the following capacities (in hours) in the planning period: A - 500, B - 480, C - 370.

The following table represents the manufacturing time (hours) per ton of product required for each type of metal sheet in the different lines: Formulate a linear programming problem to obtain

Line	Sheet 0.4 mm	Sheet 0.5 mm	Sheet 0.6 mm	Sheet 0.7 mm
A	0.4	0.3	0.3	0.25
B	0.80	0.6	0.75	0.6
C	-	0.5	0.4	-

the monthly production plan (quantity of metal sheet in tons of each type of metal sheet to be manufactured in each production line).

9. Soft drinks Spellman's Ltd is a company that produces two types of chilling soft drinks: A , and B . Both beverages use a semi-elaborate C , another expensive ingredient D and other ingredients that are not relevant for production planning.

The selling price of drink A is 3€/liter and the selling price of drink B is 2€/liter.

1 liter of drink A uses 3 grams of ingredient D . A liter of drink B uses 1 gram of ingredient D . There are only 3 grams of ingredient D available per day.

The factory only has one mixer to elaborate both drink types and the semi-elaborate. It takes 1 hour to process a liter of drink A , 1 hour to process 1 liter of drink B , and 1 hour to process 1 cl of semi-elaborate C . The mixer is available 6 hours per day.

Drink A uses 2cl of semi-elaborate C and drink B uses 1cl of semi-elaborate C . The company has 3cl of semi-elaborate C plus the amount they decide to produce every day.

The following problem model provides the optimal daily production:

Decision variables:

- x_A : Production of drink A in liters,
- x_B : Production of drink B in liters,
- x_C : Production of semi-elaborate C in centiliters.

$x_A, x_B, x_C \in \mathbb{R}$.

Objective function:

$$\max z = 3 \cdot x_A + 2 \cdot x_B$$

z is the profit in euros.

Constraints

Availability of ingredient D in grams:

$$3 \cdot x_A + x_B \leq 3.$$

Availability of mixer in hours:

$$x_A + x_B + x_C \leq 6.$$

Availability of semi-elaborate C in centiliters:

$$2 \cdot x_A + x_B - x_C \leq 3.$$

Logical constraint: $x_A, x_C, x_B \geq 0$.

A solver provides the following solution:

Optimal

Total profit is 6.00 €

The following table shows the decision variables:

j	Variables	Solution	Reduced cost	Objective Coefficient	Objective Lower bound	Objective Upper bound
A	Production A (Liters)	0	-3	3	-inf	6
B	Production B (Liters)	3	0	2	1	inf
C	Production C (Liters)	3	0	0	-0	1.5

The following table shows the constraints:

j	Constraint	Slack	Shadow Price	Right Hand Side	Min RHS	Max RHS
0	Availability of ingredient D (Liters)	0	2	3	0	4.5
1	Availability of mixer (hours)	0	0	6	3	inf
2	Availability of semi-elaborate (Liters)	3	0	3	0	inf

Answer the following analysis questions:

1. Operational problems in the mixer, like mechanical problems, could reduce the available mixer. Based on the provided solution, how is the company able to deal with this type of problem? Motivate your response.
2. The managers of the company want you to negotiate with a new supplier that could increase your daily supply of ingredient D. What would be your terms for negotiating with the new supplier, ensuring that the basic variables remain the same?
3. The managers of the company would like to review the prices of both products. Analyse how changes in the prices of both products could affect the production plan and provide useful feedback to guide the revision of the prices.

Duality

Write down the dual problem.